

Christopher Miller

301-377-3019 | chrismiller1599@gmail.com | github.com/ccm108 | chriscode.io

EDUCATION

Bowie State University

Bachelor of Science in Computer Science, Cybersecurity Focus

Bowie, MD

August 2022 – May 2026

SKILLS

Application & Cloud Security: OWASP Top 10, OWASP API Top 10, Secure SDLC, DevSecOps, CI/CD Pipeline Security, API Security, REST API Authentication, SAST, DAST, Vulnerability Assessment, CVSS, Threat Modeling, Cryptography, CIS Benchmarks, Container Security, Zero Trust, Infrastructure as Code (IaC), NIST RMF, NIST SP 800-53

Security Tools & Frameworks: Burp Suite, OWASP ZAP, Splunk, Nmap, Wireshark, Sysmon, MITRE ATT&CK, Atomic Red Team, Azure DevOps, Docker, Kubernetes

Cloud & Systems Infrastructure: AWS, Azure, Terraform, Linux (Ubuntu, Kali), Windows Server, Git, GitHub, VirtualBox

Programming & Automation: Python, Bash, PowerShell, SQL, JavaScript (React.js), HTML/CSS

EXPERIENCE

AI Research Assistant

March 2026 – May 2026

Bowie, Maryland

On-site

- Benchmarked local Llama 3 model accuracy via Ollama by developing automated Python inference scripts to execute and log a dataset of 1,801 healthcare-related queries.
- Quantified response alignment and semantic similarity by programming comparison scripts utilizing Sentence-BERT (SBERT) embedding vectors against baseline answer keys.
- Engineered data visualization dashboards using Python to plot error density heatmaps, precision-recall curves, and F1-score distributions for weekly technical reporting.

Bulldog Cybersecurity Club

August 2023 – May 2026

Bowie, Maryland

On-site

- Participated in hands-on cybersecurity labs practicing offensive and defensive techniques including network scanning, vulnerability assessment, and security tool operation across club-organized training sessions.
- Engaged in in-school CTF competitions and technical workshops studying emerging security tools and attack methodologies to build practical cybersecurity skills alongside coursework.

PROJECTS

AWS IAM Security Audit Lab | Terraform, AWS IAM, EC2, IAM Access Analyzer | GitHub | 2026

- Designed and deployed a 3-tier IAM architecture across Admin, Developer, and Auditor roles by authoring custom least-privilege policies pinning EC2 mutations to a single instance ARN, confirmed by zero privilege overlap across all roles during live testing.
- Eliminated pre-authentication access risk by implementing an account-wide RequireMFA policy using aws:MultiFactorAuthPresent conditions, verified by 100% of tested actions returning AccessDenied prior to MFA enrollment across all three IAM groups.
- Deployed a fully reproducible AWS environment by writing modular Terraform across 6 files covering IAM, EC2, networking, and Access Analyzer resources, provisioning 22 resources with zero manual console configuration and clean terraform apply output.
- Confirmed zero external exposure across all provisioned resources by running IAM Access Analyzer against the full account boundary, producing a documented 0 findings audit result with structured security rationale mapped to AWS's official IAM security guidelines.

Virtual SOC Homelab | Kali Linux, Windows, Metasploitable2, DVWA, Splunk (SIEM), Atomic Red Team, Sysmon | 2026

- Simulated offensive attack techniques against vulnerable targets using Atomic Red Team, collecting and logging resulting security events across Windows endpoints via Windows Event Logger and Splunk (SIEM).
- Documented security events and attack behavior patterns generated from simulated intrusions, analyzing Splunk alert data to validate detection coverage and identify gaps in endpoint visibility.

Automated Network Traffic Analysis Engine | Python, Scapy, tcpdump, PCAP Forensics, File I/O | GitHub | 2026

- Engineered an automated network analysis framework in Python using Scapy to ingest raw PCAP files, decoding low-level packet metadata and TCP flags to eliminate manual capture inspection.
- Collected network traffic packets in a Linux environment using tcpdump and Scapy, capturing raw packet data across simulated attack scenarios for downstream security analysis.
- Analyzed and documented network traffic using Python automation, identifying anomalous patterns including port scanning and IP flooding events and exporting structured forensic reports in JSON and CSV formats.

CERTIFICATIONS

CompTIA Security+

In Progress

Core Areas: Application Security, Threat Intelligence, Network Security, Cryptography, Vulnerability Management